DBrain: Distributed Spiking Neural Network Simulation on a Self-Organizing Wireless Cluster of Commodity Microcontrollers

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Project repository: github.com/AliAkrami1375/DBrain

Impact Statement

Research-grade neuromorphic machines are confined to a handful of laboratories; this work asks how much of their capability can be rebuilt from microcontrollers costing a few dollars each, connected only by radio. The proposed architecture turns off-the-shelf Wi-Fi microcontroller boards into a single self-organizing simulated nervous system that grows when a board is powered on and degrades gracefully when one fails. Beyond its value for research into distributed real-time simulation, the platform puts hands-on spiking neural network experimentation within reach of any classroom, laboratory, or individual, and its measurement-driven design method — deriving every architectural decision from quantified channel, memory, and timing constraints — transfers directly to other resource-constrained distributed sensing and computing systems.

Abstract

The human brain computes on roughly twenty watts; the machines built to imitate it fill server racks. Between these extremes lies an almost unexplored design point: distributing meaningful spiking neural network (SNN) simulation across commodity Wi-Fi microcontrollers connected by nothing but the air between them. This paper presents DBrain, a system architecture for simulating SNNs on self-organizing clusters of ESP32-family and ESP8266 microcontrollers communicating exclusively over ESP-NOW, a connectionless 802.11 protocol. The design is derived from primary measurements and documented constraints: an Izhikevich neuron model in s16.15 fixed-point arithmetic for bit-identical behaviour across heterogeneous chips; spike transport as address-event bundles carried by broadcast frames treated as hardware multicast; a loss policy of counting rather than retransmitting, with biological synaptic

delays absorbing radio latency; sub-millisecond clock synchronization built on hardware receive timestamps; and time-division transmission slotting enabled by that synchronization. Capacity analysis against measured channel behaviour indicates that a 40-board cluster sustains approximately 40,000 neurons and 10 million synapses at biological firing rates — about one third the neuron count of a fruit fly and two orders of magnitude beyond the complete nervous system of Caenorhabditis elegans, whose connectome constitutes a natural first milestone. No prior system, academic or otherwise, appears to have exchanged neural spikes between microcontrollers over radio. The approach is argued to be feasible, its limits are quantified, and a staged, measurement-gated implementation path is set out.

Keywords: spiking neural networks; neuromorphic computing; distributed simulation; microcontrollers; wireless networks

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1. Introduction

The most complex object known to science runs on the power budget of a dim light bulb. Eighty-six billion neurons, wired through on the order of a hundred trillion synapses, produce thought, memory, and motion while consuming about twenty watts (Herculano-Houzel, 2009). The machines built to imitate that trick occupy the opposite corner of the design space: SpiNNaker fills a machine room with a million ARM cores (Furber et al., 2014), and Intel's Hala Point packs 1,152 Loihi 2 chips drawing 2.6 kW (Davies et al., 2018; Intel, 2021). Both are remarkable, and both are physically and financially unreachable for almost everyone who might want to hold a spiking network in their hands.

This paper asks a deliberately modest question with a rich answer: how far can spike-level neural simulation be pushed on the cheapest, most abundant computers ever manufactured? The ESP32 family of Wi-Fi microcontrollers sells for a few dollars per board, with hundreds of millions of units shipped per year. Each carries one or two cores in the 100–240 MHz range, a few hundred kilobytes of RAM, and — decisively — a radio. Espressif's ESP-NOW protocol lets these chips exchange frames directly, with no access point, no TCP/IP stack, and no infrastructure: two boards powered anywhere near each other can communicate.

DBrain is a system architecture that turns an arbitrary collection of such boards into a single simulated nervous system. Each board owns a shard of neurons and synapses. When a neuron fires, the event crosses the air as part of a broadcast frame and is delivered into synapses on every subscribing board. The cluster is self-organizing: boards discover each other, elect a coordinator, agree on time, and accept a share of

the network proportional to their capability. Unplugging a board injures the brain; powering a new one grows it.

The contribution is threefold. First, a feasibility study grounded in primary sources and published measurements (Section 3): the payload, throughput, latency, and timing properties of ESP-NOW across the chip family, and the memory and arithmetic budget of each variant. Second, a complete system design (Section 4) that transfers the architectural lessons of large neuromorphic machines — source-addressed multicast spikes, logical time, drop-rather-than-retry congestion policy, event-driven synapses — onto a broadcast radio medium, together with the distributed-systems machinery a wireless cluster additionally requires. Third, an honest capacity analysis (Section 5) locating the achievable scale on the biological axis: well beyond the complete nervous system of C. elegans, within reach of insect-brain subcircuits, and eight orders of magnitude short of a human brain — a gap that is quantified rather than obscured.

A systematic search across academic literature and engineering communities found no prior system in which microcontrollers exchange neural spikes over radio (Section 2.3). DBrain therefore occupies a genuinely open point in the design space: the architecture of the room-sized machines, at three dollars per node, over the air. The design document and all project artefacts are openly available in the project repository (github.com/AliAkrami1375/DBrain).

2. Related work

2.1 Large-scale neuromorphic machines

SpiNNaker (Furber et al., 2014) remains the reference architecture for real-time SNN simulation on programmable cores, and most of DBrain's load-bearing decisions descend from it. Its million ARM968 cores each simulate on the order of a thousand simple neurons on a 1 ms wall-clock tick. A spike is a 40-bit packet carrying only the identity of the neuron that fired; router tables compiled offline replicate each packet toward every subscribing core. Three of its choices matter here. First, there is no global barrier and no global clock: cores that overrun a busy tick catch up in quiet ones (Rhodes et al., 2018). Second, congested routers drop packets rather than retry, on the grounds that biological computation tolerates sparse spike loss but not stalled real time (Rowley et al., 2019). Third, neuron state lives in fast local memory while the far larger synaptic matrix resides in off-chip SDRAM, streamed only when a spike arrives (Rhodes et al., 2018). Notably, SpiNNaker 2 moved to Cortex-M4F cores at a few hundred megahertz (Mayr et al., 2019) — silicon of the same class as the microcontrollers considered here.

Intel's Loihi advances time by barrier synchronization events rather than a clock — a timestep is a logical agreement among cores (Davies et al., 2018). IBM's TrueNorth runs a global 1 ms tick and stamps each spike packet with a small delivery-tick field so a jittery transport can still deliver events at the correct logical time (Akopyan et al., 2015). Both ideas transfer directly to a radio medium. BrainScaleS (Pehle et al., 2022) takes the opposite pole — analogue neurons running thousands of times faster than biology — and serves here mainly as contrast.

2.2 Small systems and microcontroller projects

At the opposite end of the scale, Spikeling implements one Izhikevich neuron on a single ESP32 as a teaching instrument, with units interconnected by audio cables (Chagas et al., 2018) — evidence that real-time Izhikevich dynamics are comfortable on this class of silicon. NeuroBytes produced physical printed-circuit neurons joined by cables; the project was grant-supported and crowdfunded, shipped kits, and reconstructed a flatworm connectome before ceasing activity around 2020 (NeuroTinker, 2020). Its trajectory demonstrates both the appeal of tangible neurons and the scaling ceiling of wired interconnect; the Lu.i educational neuron continues the niche in analogue form (Billaudelle et al., 2025). General-purpose ESP32 clusters have been assembled — notably the Broccoli task-queue cluster (Lin, 2018) — and abandoned once demonstrated, with the network as bottleneck and no compelling workload. Recent work has partitioned conventional (non-spiking) neural network inference layer-wise across wireless microcontrollers (Sensors, 2025), the nearest published relative of the present proposal.

2.3 The gap

Searches across arXiv, IEEE, ACM, GitHub, and engineering communities found single-chip neuron instruments, wired neuron boards, generic microcontroller clusters, and rack-scale neuromorphic machines — but no system exchanging spikes between microcontrollers over any radio protocol. The documented failure modes of the neighbours are instructive: flooding meshes collapse under dense traffic; per-message textual serialization drowns in overhead; central-master orchestration without a compelling demonstration precedes abandonment. The design below is shaped accordingly.

3. Background and constraints

Every architectural decision in Section 4 is forced by a measured or documented constraint; this section fixes those constraints.

3.1 The transport

ESP-NOW transmits vendor-specific IEEE 802.11 action frames, connectionless and infrastructure-free (Espressif Systems, 2025a). The version 1 payload limit — and the limit for any frame an ESP8266 must receive — is 250 bytes; version 2 extends this to 1,470 bytes on newer chips, but version 1 devices cannot receive larger frames. The peer table holds at most 20 entries, yet broadcast frames require no peer registration and reach an unlimited audience — the property on which DBrain's transport rests. Broadcast frames receive no link-layer acknowledgement and no retransmission; unicast frames are acknowledged and retried, though a receiver whose buffer overflowed still acknowledges, so link-layer success does not guarantee application delivery (MicroPython Project, 2024).

Measured behaviour fixes the performance envelope. At the default 1 Mbps physical rate, independent benchmarks report roughly 90–100 kB/s of application throughput and approximately 345–404 maximum-size frames per second on a saturated channel (NTU PS9888 Group, 2020); raising the rate to 24–54 Mbps (unavailable to the ESP8266) yields 780–960 kB/s for broadcast. End-to-end latency for small frames has a median of 5.6 ms including the software path (Electric UI, 2024). Two timing measurements are architecturally decisive: a single broadcast reaches multiple receivers within a spread of only about 80 µs, while the sender's own transmit instant jitters by up to 5 ms under carrier-sense contention and task scheduling (Arduino Forum, 2024). Synchronization must therefore be built on reception, never transmission. Recent versions of the vendor framework expose a microsecond-resolution hardware receive timestamp in the receive callback (Espressif Systems, 2025a), providing exactly the required primitive; the 802.11 TSF timer, by contrast, does not run without an access-point association and is unusable here.

3.2 The chip family

Verified against vendor documentation (Espressif Systems, 2025b): the ESP32, S2, S3, C2, C3, C5, and C6 all carry 2.4 GHz Wi-Fi and support ESP-NOW, with 272–520 KB of SRAM and one or two cores at 120–240 MHz. The ESP8266 participates with version 1 frames only, roughly 40–50 KB of usable heap, and no hardware receive timestamp. Two family members are excluded by physics rather than software: the H2 has no Wi-Fi radio, and the P4 has no radio at all. Critically, most of the family — including, counterintuitively, the S2 — has no floating-point unit; on FPU-less chips a software float addition costs roughly 100 cycles and a cosine roughly 2,400 (Espressif Systems, 2025c). Optional external PSRAM (4–8 MB on common modules) is an order of magnitude slower than internal SRAM but tolerates sequential streaming well.

3.3 Neuron model economics

Izhikevich's canonical comparison (Izhikevich, 2004) prices the candidate models: a leaky integrate-and-fire neuron costs about 5 operations per simulated millisecond but reproduces only 3 of roughly 20 catalogued biological firing behaviours; the Hodgkin–Huxley model reproduces all of them at about 1,200 operations; the Izhikevich model reproduces all 20 at 13 operations with two state variables. Cortical firing rates follow a lognormal distribution spanning roughly 0.1–10 Hz with means of a few hertz (Buzsáki and Mizuseki, 2014) — a sparsity that makes event-driven synaptic processing, and by extension radio-linked simulation, feasible at all.

4. System design

4.1 Simulation core

Each node runs the hybrid loop on which real-time SNN engines have converged (Rhodes et al., 2018): time-driven neuron integration on a fixed tick (1 ms of biological time), with synaptic work performed only on spike arrival. Incoming spikes index a per-source synaptic row; each traversed synapse adds its weight into the target neuron's input accumulator at the tick indicated by its delay, implemented as a ring of per-tick buckets.

All arithmetic is integer fixed-point in the s16.15 format — the format used, with published accuracy analyses, throughout SpiNNaker's production software (Hopkins and Furber, 2015; Rhodes et al., 2018). The choice is forced twice over: most supported chips have no FPU (Section 3.2), and a heterogeneous cluster must produce bit-identical results wherever a neuron is placed, so that placement is invisible to the simulation and any node's behaviour can be reproduced exactly on a host-side reference simulator. Synapses pack into 32 bits — 16-bit weight, 4-bit delay, 8-bit local target index, type and padding — following SpiNNaker's layout. Neuron state occupies 8 bytes. On a board with 4 MB of PSRAM, neuron state resides in internal SRAM while synaptic rows stream from PSRAM on spike arrival, mirroring SpiNNaker's SDRAM split; the budget supports on the order of 1,000 neurons and one million synapses per board (Section 5).

4.2 Spike transport

Spikes cross board boundaries in address-event representation (Boahen, 2000): a packet identifies which neuron fired and nothing else. DBrain's key observation is that an ESP-NOW broadcast is hardware multicast: one transmitted frame is received by every node in the cell. Where SpiNNaker replicates packets through compiled router

tables, DBrain broadcasts each spike bundle once; each receiving node binary-searches the source identifier against a local subscription table — the software analogue of SpiNNaker's ternary content-addressable routing memory — and either processes or discards the bundle. There are no routing tables, no forwarding, no per-peer state, and no dependence on the 20-peer limit. A spike consumed by seven boards costs one frame.

Fired spikes accumulate during a tick and transmit as a single bundle of up to 119 two-byte events in a 250-byte frame, under a 12-byte header carrying the source node, tick number, and a per-node sequence number. Version 1 framing is retained for all simulation traffic so that every supported chip can receive it.

The loss policy is imported deliberately from SpiNNaker (Rowley et al., 2019): spike bundles are fire-and-forget; sequence-number gaps are counted and reported, never repaired. A retransmitted spike is a late spike, and a late spike is either wrong or a stall. Two mechanisms make this safe. First, biological synaptic delays (1–15 ms; the 4-bit delay field) absorb transport latency exactly as TrueNorth's delivery-tick field does (Akopyan et al., 2015): a bundle stamped tick T arriving during tick T+1 still applies every synapse of delay ≥ 2 at its correct biological time. The network compiler enforces a minimum cross-board delay of two ticks, rendering ordinary radio jitter invisible to the simulation. Second, control-plane traffic — join, assignment, migration — uses the opposite contract: application-level acknowledgement with bounded retry over unicast.

4.3 Time

Logical time advances in ticks; the wall-clock duration of one tick is a cluster constant (the slowdown factor) chosen from the airtime budget — unity for small, fast clusters; five to ten for large or ESP8266-inclusive ones. There is no per-tick barrier: as on SpiNNaker, an overrunning node catches up during quieter ticks, and the delay slack of Section 4.2 hides transient lag. A paced mode with explicit per-tick acknowledgement exists for hard-reproducibility experiments at the price of running at the speed of the slowest node.

Synchronization exploits the measured asymmetry of Section 3.1. The coordinator broadcasts a beacon each second carrying the tick count and its own clock; every receiver pairs the beacon with its hardware receive timestamp and maintains an eight-point linear regression of coordinator time against local time, slewing — never stepping — its tick phase. Crystal tolerances of ±10 ppm (Espressif Systems, 2025d) bound inter-node drift at 20 µs/s, so residual error remains far inside a tick between beacons, and the regression suppresses crystal skew to the ~1 ppm class, as in the flooding time

synchronization protocol of Maróti et al. (2004). The realistic accuracy floor is set by the ~80 µs receiver-side spread; the design targets under 100 µs cluster-wide, noting candidly that published claims of tighter synchronization over this transport lack rigorous measurement. One hundred microseconds is one tenth of a tick — sufficient with margin for the mechanism synchronization exists to serve: once nodes share time, each tick's transmission window is divided into assigned time-division slots, converting synchronized burst collisions — the broadcast channel's worst enemy, unprotected by acknowledgements and aggravated by hidden nodes — into scheduled, collision-free airtime. Synchronization enables slotting; slotting protects the beacons that maintain synchronization. ESP8266 nodes, lacking the hardware timestamp, synchronize at millisecond precision in software and receive proportionally wider guard bands.

4.4 Self-organization

Membership derives entirely from once-per-second jittered broadcast heartbeats carrying each node's identity, capability record (chip, cores, clock, free memory, PSRAM, FPU), epoch, and health counters. On a broadcast medium every node overhears every heartbeat, so passive listening replaces most of an active gossip protocol; a hundred-node table costs about 2.4 KB. Failure detection follows the SWIM pattern (Das et al., 2002): missed heartbeats mark a node suspect, indirect probes through two peers guard against asymmetric radio links, and confirmed death triggers reassignment. A new board broadcasts join beacons across channels until it hears the cluster, then receives its identity and, if resources allow, a shard.

The coordinator is deliberately not a master: the spike plane is peer-to-peer and continues — degraded — with no coordinator at all. Its duties are the slow path only: beacons, admission, placement, epoch numbering. Election is therefore the simplest mechanism that works, essentially the root election of Maróti et al. (2004): the highest key — capability score concatenated with hardware address — wins; a node hearing no beacon for a timeout self-promotes with an incremented epoch; higher keys and epochs pre-empt immediately. Partitions heal deterministically on merge, and split-brain during a partition costs only counted spike loss across the split. Raft-class consensus was evaluated and rejected: it presumes reliable ordered unicast and purchases a replicated log that a distinguished beacon source plus an epoch counter renders unnecessary (Rahman, 2019).

Assigning neurons to boards is graph partitioning under a multicast cost model: an off-board source neuron costs one expected frame per firing regardless of how many synapses subscribe on a given board, so the objective counts unique source–board pairs weighted by firing rate — a hypergraph formulation. SpiNeMap demonstrated 45%

energy reductions on neuromorphic hardware by partitioning against measured rather than assumed spike rates (Balaji et al., 2020); DBrain adopts the same measure-and-refine loop through its telemetry. Shard sizing balances tick compute time, not neuron count, against each node's advertised capability, honouring the bulk-synchronous rule that the slowest node sets the pace (Valiant, 1990). A compiler-less mode also exists: the coordinator broadcasts a generative recipe — topology family, parameters, seed — from which every node derives its own population and connectivity deterministically; the network is a pure function of the recipe, with no tables shipped at all.

5. Capacity analysis and biological scale

5.1 Airtime

At 1 Mbps, a 250-byte broadcast occupies roughly 2.5–2.9 ms of air including preamble, inter-frame spacing, and mean backoff — a saturation ceiling near 345 frames/s, consistent with the 404 frames/s measured under ideal conditions (NTU PS9888 Group, 2020). Holding channel occupancy at a conservative 30% (broadcast collisions being unrecoverable), the cluster-wide budget is approximately 100 bundles/s at 1 Mbps and 600–1,150 bundles/s at 24–54 Mbps — that is, roughly 12,000 and 70,000–140,000 spike events per second respectively at 119 events per bundle.

Demand scales as nodes × neurons per node × mean rate. Forty boards of 1,000 neurons at a 3 Hz mean generate 120,000 events/s — at the upper edge of the high-rate budget, which time-division slotting and rate-aware placement are designed to exploit fully. At a 1 Hz mean, well within the biological lognormal range, the same budget carries on the order of a hundred boards. With ESP8266 nodes present, the 1 Mbps compatibility contract applies and the cluster compensates with its slowdown factor — precisely the accommodation SpiNNaker's cortical-microcircuit work made in its first release before later optimization reached real time (van Albada et al., 2018).

5.2 Position on the biological axis

The comparison below uses accepted reference values: 86×10^9 neurons for the human brain (Herculano-Houzel, 2009); the complete adult Drosophila connectome of 139,255 neurons and 54.5 million synapses (Dorkenwald et al., 2024); and the 302-neuron, ~7,600-connection nervous system of C. elegans (White et al., 1986; Cook et al., 2019).

System	Neurons	Synapses	Power
Human brain	8.6×10^{10}	$\sim 10^{14}$–10^{15}	~20 W
Fruit fly (adult, complete)	1.39×10^5	5.45×10^7	$\sim 10^{-4}$ W

System	Neurons	Synapses	Power
C. elegans	3.02×10^2	$\sim 7.6 \times 10^3$	—
One SpiNNaker chip	$\sim 10^4$	$\sim 10^6$–10^7	~1 W
DBrain, 40 boards	$\sim 4 \times 10^4$	$\sim 10^7$	~20–40 W

Three readings of this table frame the project honestly. Against the human brain, the gap is a factor of about two million in neurons and ten million in synapses; closing it with this hardware would require some 86 million boards, a quarter of a billion dollars of silicon, and tens of megawatts against the brain's twenty watts. No incremental engineering closes that gap; it is stated as a boundary, not a goal. Against the small end of biology, the position is far more interesting: a 40-board cluster exceeds the complete nervous system of C. elegans — an animal that locomotes, forages, avoids, and learns with 302 neurons — by two orders of magnitude. The full worm connectome fits on two to three boards with capacity to spare, making it the natural first biological milestone: a physically distributed, radio-linked execution of a real organism's entire neural wiring, driving robotic behaviour. At the cluster's upper end, 40,000 neurons is roughly 30% of a fruit fly's brain — the scale of well-characterized insect subcircuits such as the olfactory pathway. Against the machines, a 40-board cluster is approximately one SpiNNaker chip in neuron count: the honest exchange rate between a wired backplane and the open air, purchased at two orders of magnitude lower cost, with no infrastructure, growing by powering another node.

6. Discussion and limitations

The channel is the machine: every scaling property reduces to airtime arithmetic. The mitigations — batching, time-division slotting, multicast-aware placement, the slowdown factor — are in the design, but a 2.4 GHz environment shared with domestic Wi-Fi erodes the budget unpredictably; the design makes this erosion visible through telemetry rather than pretending to prevent it. The synchronization floor is asserted, not yet measured: the sub-100 µs target rests on the documented behaviour of hardware receive timestamps and one published receiver-spread measurement, and the implementation plan places a rigorous logic-analyser measurement before any dependent component. Heterogeneity has a price: ESP8266 participation constrains audible frames to 250 bytes at 1 Mbps and lowers the whole cluster's ceiling, a cost the architecture confines to clusters that actually include such nodes. Security is deferred by design: broadcast frames cannot be encrypted at the protocol layer, so the first version is an open laboratory instrument, with an application-layer authenticated-encryption path specified but unimplemented. Finally, spike loss is counted, bounded,

and biologically defensible — but it is loss; experiments requiring exact reproducibility must use the paced mode or replay on the bit-identical host reference simulator.

7. Roadmap and future work

Implementation proceeds in measurement-gated phases, each ending in a running artefact and a measured number: a transport benchmark suite replacing this paper's borrowed channel figures with fleet measurements; two boards exchanging a synfire chain over the air; synchronization and slotting with the residual measured by logic analyser; live membership under a running raster; the network compiler and bit-identical host reference; a gateway and browser dashboard with live rasters and telemetry; and finally spike-timing-dependent plasticity, whose deferred event-driven formulation transfers from SpiNNaker unchanged because presynaptic spikes arrive at the board that owns the synapse (Diehl and Cook, 2014). The biological milestone — the complete C. elegans connectome driving a wheeled robot — follows the compiler phase. All artefacts are published in the project repository as phases complete.

8. Conclusion

The large neuromorphic machines answered whether spiking brains can be simulated in real time on programmable silicon. DBrain asks the complementary question: how much of that answer survives when the backplane is replaced by air and the budget by pocket money. The analysis here says: enough to matter. The protocol limits, arithmetic, memory, and measured timing of commodity Wi-Fi microcontrollers jointly admit a self-organizing, radio-linked spiking network of tens of thousands of neurons — beyond a complete biological nervous system, within sight of insect subcircuits, and honest about everything it is not. The remaining questions are empirical rather than conceptual, and the roadmap is built to answer them in order.

Data availability statement

The project's design documentation, and the firmware, network compiler, and measurement artefacts as they are produced through the roadmap of Section 7, are available in the public project repository at github.com/AliAkrami1375/DBrain. The capacity analyses in this paper are reproducible from the constants and sources cited in Sections 3 and 5.

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The author declares none.

Author contributions

A.A.: Conceptualization; Methodology; Formal analysis; Investigation; Writing — original draft; Writing — review and editing.

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